Report

Current Biology

Solar heating may explain extreme diel flight altitude changes in migrating birds

Highlights

- Migrating birds are warmer at day than at night when flying at the same altitudes
- This indicates that their heat balance is affected by solar radiation at daytime
- The birds follow a diel altitude cycle, flying much higher at day than at night
- By high and cold diurnal flights, solar heating seems to be counteracted

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In brief

Great reed warblers and great snipes exhibit a diel cycle in migratory flight altitudes, flying much higher during the day than the night. Using logger temperature data, Sjöberg et al. show that birds at the same altitude are warmer during the day than at night, indicating that they climb to cold altitudes to counterbalance solar heating in daytime flights.

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Current Biology

Report

Solar heating may explain extreme diel flight altitude changes in migrating birds

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SUMMARY

Great reed warblers, *Acrocephalus arundinaceus*,[1] and great snipes, *Gallinago media*,[2] exhibit a diel cycle in flight altitudes—flying much higher during the day than the night—when performing migratory flights covering both night and day. One hypothesis proposed to explain this behavior is that the birds face additional heating by solar radiation during daytime and hence must climb to very high, and thus also very cold, altitudes to avoid overheating during daytime flights.[1,2] Yet, solar heat gain in birds has been shown to drastically decrease with wind speed,[3,4] and the quantitative heating effect by solar radiation on a bird flying with an airspeed of 10 m/s or more is unknown. We analyzed temperature data from multisensor data loggers (MDLs)[5,6] placed without direct exposure to solar radiation on great reed warblers (the logger covered by feathers on the back) and great snipes (the logger on the leg, covered from the sun by the tail). We found that logger temperatures were significantly higher (5.9°C–8.8°C in great reed warblers and 4.8°C–5.4°C in great snipes) during the day than during the night in birds flying at the same altitudes (and thus also the same expected ambient air temperatures). These results strongly indicate that the heat balance of the flying birds is indeed affected by solar radiation, which is in accordance with the hypothesis that solar radiation is a key factor causing the remarkable diel cycles in flight altitude observed in these two long-distance migrant bird species.[1,2]

RESULTS AND DISCUSSION

Great reed warblers ("warblers") and great snipes ("snipes") were equipped with multisensor data loggers (MDLs) recording light intensity, temperature, barometric pressure (altitude), and accelerometer data (flight activity).[5,6] The logger data revealed an unexpected and extreme migratory flight behavior, where the two species showed a pronounced and remarkable diel cycle in flight altitudes, flying 2,000–4,000 m higher during the day than during the night[1,2] (Figure 1). In this paper, we analyze the logger temperature data to investigate the possibility that solar radiation is an explanatory factor for the birds' regular and dramatic changes in flight altitude between night and day.

In the height interval 2,000–5,000 m above sea level (asl), the logger temperature at a given flight altitude was 5.9°C–8.8°C, and 4.8°C–5.4°C higher during the day than the night for the warblers and the snipes, respectively (Tables 1 and 2; Figure 2). We focused on this height interval because flights occurred there both during day and night, allowing direct comparison.

Air density, ambient air temperature, and oxygen pressure are known to affect birds' flight performance and become markedly reduced with increasing altitude.[7] Similarly, wind conditions often differ considerably between altitudes. However, at heights exceeding ~2,000 m asl, these conditions show no systematic changes at a given altitude between night and day,[8,9] thus making it unlikely that any of these factors are primary causes for the observed regular diel cycle in flight altitude. The only factor we can identify that differs markedly between day and night at the same altitude is the possible heating by solar radiation of the birds' body and plumage during daytime, and this may affect logger temperature. The data presented in Figure 2 and Table 2, showing significant heating of the air inside the plumage of the birds during daytime (as measured by the data loggers shielded from direct sunlight), are in accordance with the hypothesis that birds ascend to fly at much higher and colder altitudes during the day compared with the night to avoid overheating from solar radiation at daytime.[1,2]

Potential heat/energy gain from solar radiation

The "heat balance" hypothesis[1,2] was inspired by studies showing that the flight activity of bats during the day may be constrained by hyperthermia caused by the exposure to solar radiation.[10,11] Solar radiation has also been suggested to be a potentially significant contributor to the heat balance of flying birds.[3,12,13] Moreover, flight duration of migrating common eiders, *Somateria mollissima*, has been demonstrated to be constrained by hyperthermia,[14] which may be aggravated by the fact that these ducks are diurnal migrants and thus exposed to solar radiation.

We estimate that the body area exposed to direct solar radiation in sunny weather would be around 20 and 50 cm^2 for a flying

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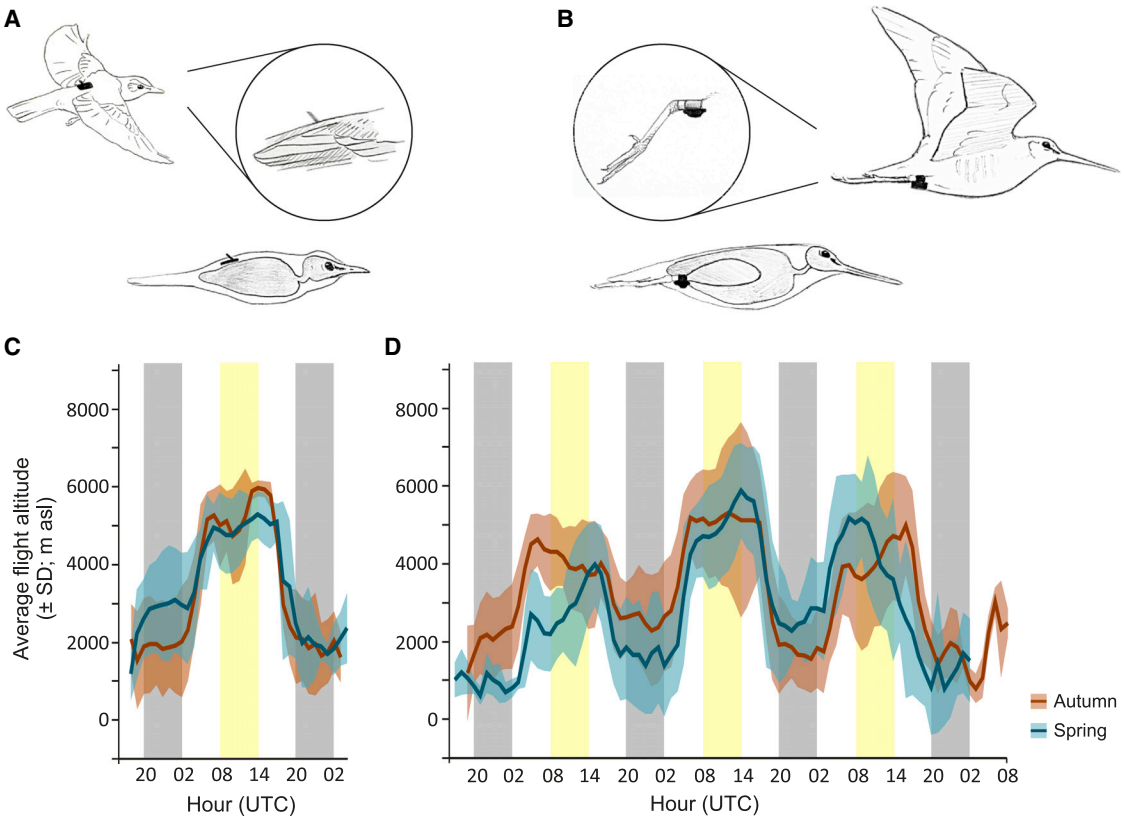


Figure 1. Data logger placement on the different species and the diel cycle in flight altitudes

(A and B) The temperature data collected for the great reed warblers (A) indicate the temperature within the plumage layer on the back of the birds, where the logger is located close to the skin and covered by feathers. In contrast, the great snipes (B) wore the logger on a ring on their tibia. During flights the great snipes most commonly stretch their legs backward under the tail feathers, resulting in the logger becoming partly covered by the birds undertail coverts but shaded from the sun by the body and tail of the bird.

(C and D) The flight altitude profiles in relation to the time of the day in great reed warblers (C; based on 12 autumn and 17 spring flights from 14 individuals) and great snipes (D; based on 13 autumn and 5 spring flights from 14 individuals), respectively. Flight altitude follows a diel cycle with extremely high diurnal and comparably lower nocturnal flight altitudes, both during autumn (brown) and spring (blue) migration (mean ± SD; yellow and gray bars indicate parts of the flights that are included in the analyses as diurnal, 08–14 UTC, and nocturnal, 20–02 UTC, respectively; from Sjöberg et al.[1] and Lindström et al.[2]).

Bird illustrations: Axel Thorenfeldt.

great reed warbler and great snipe, respectively. With a solar energy flux at the Earth's surface of about 1,000 W/m^2, the potential energy/heat gain from solar radiation would thus be 2 and 5 W for flying individuals of these two species, respectively. This is of the same order of magnitude as the total metabolic flight power, being approximately 2.5 and 12 W, respectively, for a bird with a body mass of about 25 g (great reed warbler) and 150 g (great snipe) according to allometric relationships of empirical power estimates for birds in flapping flight.[15] This illustrates the great potential impact of solar radiation. Still, it could not be taken for granted that solar radiation has a significant effect on the heat balance of a flying bird. The effective heat gain is very sensitive to the structure and behavioral adjustment of the bird's feather plumage (which is a thermal buffer between the body and the environment), as well as to the flow of air around the bird's body (which depends on wind speed or flight speed).[6] For example, the experiments by Wolf and Walsberg[3,4] indicate that the solar heating significantly decreases with increasing wind speed around the bird's body. Hence, for birds flying through the surrounding air with speeds about 10 m/s or more (approximately the air speeds of flying birds[16]), convection could have reduced the heating effect of the solar radiation to low and insignificant levels.

Predictions about logger temperatures

We expected the logger temperatures to mainly reflect the temperature of the air inside the plumage. This in turn depends on body/leg temperature, ambient atmospheric air temperature, and the exchange of air through the plumage. We presumed that the temperature sensors were relatively more influenced by ambient air temperature in the snipes since the leg-mounted loggers were more exposed to the open air and further away from the core body (Figure 1). In contrast, in the warblers the loggers would be more influenced by body temperature and muscle-power-generated heat (Figure 1). Crucially, however, we did not expect that solar radiation directly affected the temperature of the logger, as the logger in both species was shielded from direct solar radiation.

Following the heat balance hypothesis, we could make a number of predictions regarding logger temperature if the birds are

Table 1. Descriptive statistics for flight altitudes, expected ambient air temperatures (according to the standard atmospheric model), logger temperature, the slope for the fitted models (logger temperature ~ flight altitude * day vs. night + 1|individual + 1|day/night number; only data between 2,000 and 5,000 m included), and the calculated difference in logger temperature between day and night at 2,000 and 5,000 m asl, for great reed warblers and great snipes, respectively

Species	Period	Flight altitude (mean ± SD; m asl)	Expected air temperature (mean ± SD; °C)	Logger temperature (mean ± SD; °C)	Slope logger temperature vs. flight altitude (2,000–5,000 m; °C/m)	Difference in logger temperature between day and night (°C)
Great reed warbler	day	5,029 ± 934	−17.7 ± 6.1	27.9 ± 4.0	−0.0039	8.8 at 2,000 m 5.9 at 5,000 m
	night	2,433 ± 1,249	−0.8 ± 8.1	27.5 ± 4.2	−0.0030	
Great snipe	day	4,264 ± 1,592	−12.7 ± 10.3	13.5 ± 8.2	−0.0050	5.4 at 2,000 m 4.8 at 5,000 m
	night	1,893 ± 1,099	2.7 ± 7.1	20.4 ± 8.2	−0.0049	

heated by solar radiation during diurnal flights. These predictions are dependent on the differences in logger placement between the two study species (Figure 1). First, ambient air temperature decreases by 6.5°C per kilometer of increasing altitude in the standard atmosphere. Thus, a negative relationship between logger temperature and flight altitude within the day as well as the night was predicted for both species. Second, the slopes of temperature vs. altitude were predicted to be steeper in the snipes compared with the warblers, depending on the difference in the relative influence of ambient and body temperature as described above (Figure 1). As predicted, logger temperatures differed between the two species, both in general level (intercept) and in relation to altitude (slope; Table 2; Figure 2). The temperatures recorded were intermediate between estimated ambient air temperature and body temperature, with body temperature (normally ca. 40°C for both species[17]) having a larger relative influence on the measured logger temperatures for the warblers than for the snipes.

By comparing logger temperatures at the same altitudes between night and day, we can control for differences in ambient air temperature and increased workload at different altitudes, and the only factor that may differ is the potential effect of extra heating of the body and the within-plumage air layer caused by solar radiation during daytime. Thus, if there are no differences in logger temperature levels between day and night (i.e., with and without solar radiation) at the same flight altitudes, this would argue against solar radiation as a key explanation for the dramatic diel flight altitude cycles in warblers and snipes. In contrast, the critical prediction of the heat balance hypothesis is that heating from solar radiation would be manifested in indirect heating of the logger, resulting in significantly higher logger temperatures during daytime than at night. This latter prediction is clearly

Table 2. GLMM model output for the effect of day vs. night and flight altitude on logger temperature for great reed warblers and great snipes, respectively

Great reed warbler—logger temperature (dependent variable)[a]					
Fixed effects:	**Estimate**	**SE**	**Df**	**t**	**P**
Intercept	44.71	2.49	15.38	17.93	< 0.001
Altitude	−0.0039	0.00044	177.2	−8.83	< 0.001
Day vs. night (night)	−10.72	2.01	176.8	−5.34	< 0.001
Altitude * day vs. night (night)	0.00097	0.00050	176.5	1.92	n.s.
Random effects:	**Variance**	**SD**			
Individual (intercept)	16.58	4.07	N/A	N/A	N/A
Day number (intercept)	2.91	1.70	N/A	N/A	N/A
Residual	4.21	2.05	N/A	N/A	N/A
Great snipe—logger temperature (dependent variable)[a]					
Fixed effects:	**Estimate**	**SE**	**Df**	**t**	**P**
Intercept	33.87	2.00	17.51	16.95	< 0.001
Altitude	−0.0050	0.00034	431.0	−14.97	< 0.001
Day vs. night (night)	−5.74	1.74	426.9	−3.30	<0.01
Altitude * day vs. night (night)	−0.00018	0.00052	427.3	0.34	n.s.
Random effects:	**Variance**	**SD**			
Individual (intercept)	10.94	3.31	N/A	N/A	N/A
Day number (intercept)	6.25	2.50	N/A	N/A	N/A
Residual	15.15	3.89	N/A	N/A	N/A

[a]Only data between 2,000 and 5,000 m asl is included in the models, see STAR Methods.

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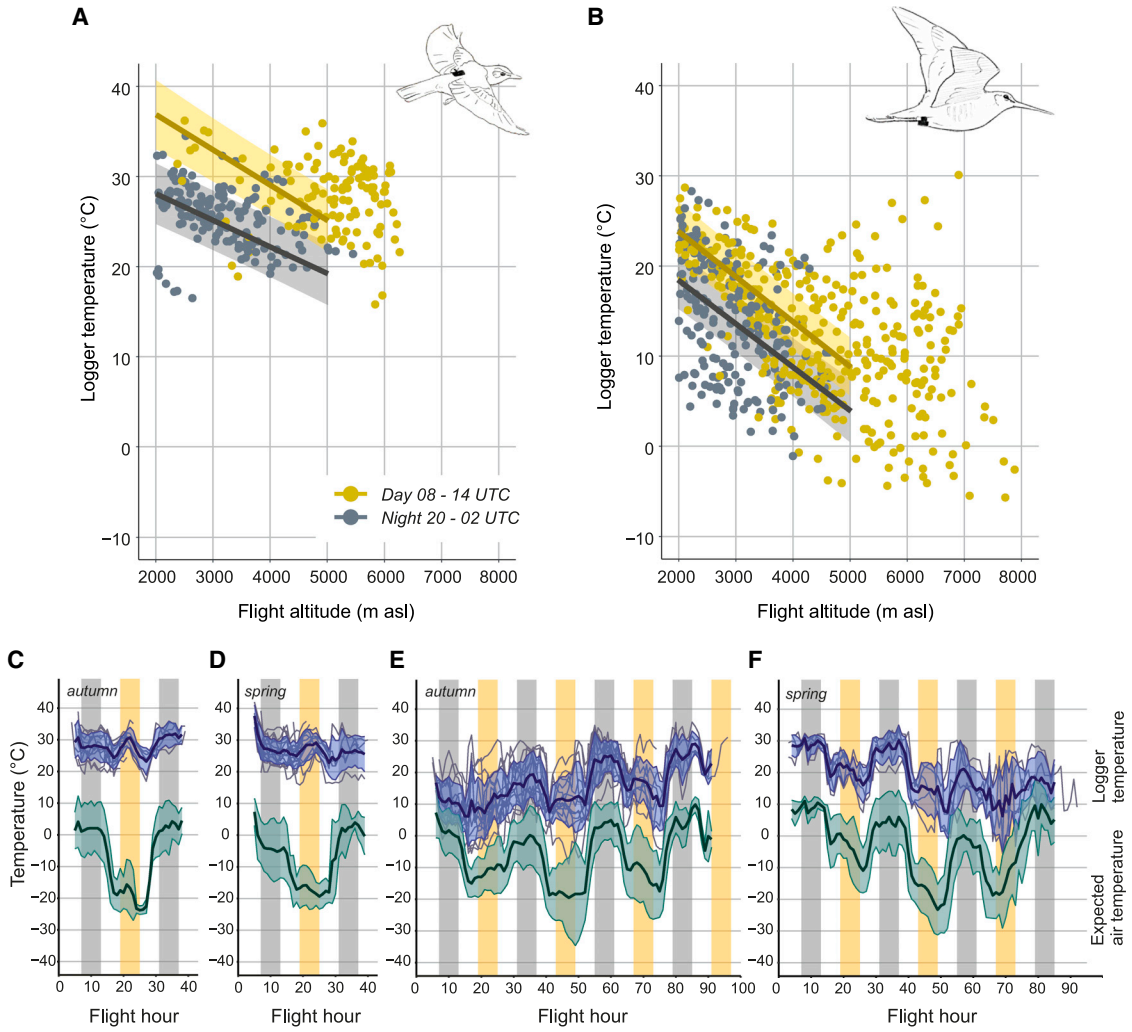


Figure 2. Difference in logger temperature between the diurnal and nocturnal parts of the flights

(A and B) For a given altitude, logger temperature was higher during diurnal flights (yellow) compared with nocturnal flights (gray) in both great reed warblers (A) and great snipes (B). Points indicate the recorded logger temperatures and flight altitudes, lines and shadowed areas illustrate linear regressions of logger temperature in relation to flight altitude ± CI as described by the fitted models; only data between 2,000 and 5,000 m is included in the models, data below 2,000 m is excluded from this figure as we cannot predict the effect of differences of ambient temperature between night and day on low altitudes on the logger temperatures; see STAR Methods.

(C–F) Logger temperature (gray lines, individual flights; purple, mean ± SD) varied less in relation to expected ambient air temperature (green, mean ± SD; calculated based on International Standard Atmosphere model [SA] altitudes, see STAR Methods) in great reed warblers (C)–(D) compared with great snipes (E)–(F), most probably as an effect of logger placement (yellow and gray bars indicate parts of the flights that are included in the analyses as diurnal, 8–14 UTC, and nocturnal, 20–02 UTC, respectively).

Bird illustrations: Axel Thorenfeldt.

supported by the results we found in both the warblers and the snipes (Table 2; Figure 2).

Heat balance at low and high altitudes

The warblers typically increased their flight altitude during daytime with two or three kilometers (Table 1; Figure 1), resulting in an estimated ambient air temperature drop of, on average, −17°C (Table 1). Despite this, the logger temperatures were very similar between night and day; mean logger temperatures were 27.9 (day) and 27.5°C (night; Table 1; Figure 2). This indicates that the birds effectively counteracted the heating effect of solar radiation by climbing to very high and cold altitudes, and thereby balanced the extra solar heating with the chilling effects of the low ambient air temperature at extremely high altitudes. For the snipes, the daytime increase in altitude resulted in an estimated change in ambient temperature of on average −15°C, and the logger temperature was on average 7°C lower during the day than during the night (Table 1). The lower diurnal temperatures might suggest that the heating effect of solar radiation was "over-compensated" by the birds' climb to high and cold altitudes. Yet, this "over-compensation" does not necessarily reflect temperature conditions close to the main body, which may well have remained in balance as the logger on the leg is considerably more open to the chilling effect from the

ambient air compared with the logger placement on the great reed warblers (Figure 1). In addition, some surprisingly high logger temperatures were recorded in the great snipes during diurnal flights at high altitudes (Figure 2). Several wader species, including common snipe, *Gallinago gallinago*, have been observed to sometimes fold their legs to be held inside the plumage when flying in cold conditions.[18] This behavior might explain some of the highest logger temperatures recorded at daytime and at very high and cold altitudes.

Energy and heat budget of high-altitude bird flight

In the previous studies, increased vision range and the threat of predation during daytime were suggested as alternative or contributory explanations for the diel cycles in flight altitudes.[1,2] These alternative hypotheses cannot be excluded and remain to be tested. In this paper we compared temperature records from MDLs carried by great reed warblers and great snipes to make a qualitative test of the hypothesis that the birds fly at higher altitudes during daytime to mitigate the effect of extra heating from solar radiation. Unfortunately, present knowledge is insufficient to permit any full predictive modeling of the heat balance of a flying bird (considering, e.g., the fragmentary knowledge of the complex effects of plumage and wind/flight speed on solar heat gain).

Light colors (high albedo) absorb less solar radiation, and a recent study reported that bird species migrating long distances had on average a lighter plumage than residents and short-distance migrants.[19] Although the between-species variation in coloration and migratory status is huge, this finding gives further support to the importance of solar radiation for the thermoregulation of flying birds. Yet, the heat budget of the birds will change in many ways when they climb to fly at extremely high altitudes during the daytime. Apart from the exposure to heating from solar radiation during the daytime and the increased evaporative and convective heat loss at the low ambient temperatures, additional factors will influence the heat budget during these high-altitude flights. The intensity of direct solar radiation is reduced by absorption and reflectance in the troposphere. This means that direct solar clear-sky irradiance is expected to increase by about 11% between 2,000 and 5,000 m altitude (example case, Oumbe and Wald,[20] Bumthaler et al.[21]), thus causing increased heating at higher altitudes. In addition, more muscle work will be required to fly at the low air densities at higher altitudes, with flight power being approximately proportional to $\rho^{-1/2}$ (ρ = air density; for an ideal bird *sensu* Pennycuick[22]), given that the birds select the same significant/optimal flight speed (e.g., the minimum power speed or the maximum range speed) at different altitudes. Hence, flight power is expected to increase by 5%–6% per 1,000 m increasing altitude, becoming 17% higher at 5,000 m compared with 2,000 m (the birds will also be expected to fly at higher speeds at higher altitudes, since the optimal speeds will change with air density in a similar way as flight power).[22] Provided conversion efficiency[23] remains rather constant, the increase in mechanical flight power at high altitude will lead to a corresponding increase in metabolic power and heat production from the muscle work. The increased cooling effect at the low ambient temperatures prevailing at high altitudes is presumably sufficient to compensate for the above-mentioned different contributions to heat gain associated with high-altitude daytime flights. In addition, evaporative cooling will be facilitated at high altitudes because of the generally lower relative humidity at higher altitudes.[24] To fully understand the heat balance of flying birds and, thus, reveal which factors set the limitations of bird flight, will require experimental studies of, for example, birds flying in wind tunnels with and without exposure to radiation as well as further biologging studies monitoring the incidence of solar radiation reaching the bird and the associated changes in flight physiology, performance, and behavior.

In the ancient Greek mythology, Icarus, who was wearing wings glued with beeswax, became reckless and despite his father's warnings flew so high that the sun melted the beeswax and he fell to his death. In contrast, the birds we have studied (with wings firmly attached to their bodies) seem to be experiencing an "inverted Icarus effect" as they may instead *avoid* overheating from solar radiation by climbing to very high, and thus, very cold altitudes.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- RESOURCE AVAILABILITY
  - Lead contact
  - Materials availability
  - Data and code availability
- EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS
  - Study systems
- METHOD DETAILS
  - Multisensor data loggers (MDLs)
- QUANTIFICATION AND STATISTICAL ANALYSIS
  - Data treatment and statistical analysis

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AUTHOR CONTRIBUTIONS

S.S., D.H., Å.L., and T.A. conceived and designed the study. A.A. and J.B. designed and built the multisensor data loggers. S.S., B.H., G.M., M.T., D.H., and Å.L carried out fieldwork. S.S., D.H., Å.L., and T.A. analyzed and interpreted the data. The manuscript was drafted by S.S. with input from D.H., Å.L., and T.A. All authors read and commented on the manuscript and approved the final version.

DECLARATION OF INTERESTS

The authors declare no conflict of interest.

INCLUSION AND DIVERSITY

We support inclusive, diverse, and equitable conduct of research.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Flight altitude and logger temperature data for all flights covering both night and day for the great reed warblers and great snipes included in the study.	This paper	Mendeley data: https://doi.org/10.17632/k7m3gnswdj.1
Experimental models: Organisms/strains		
Great reed warblers, *Acrocephalus arundinaceus*	wild	N/A
Great snipes, *Gallinago media*	wild	N/A
Software and algorithms		
R	Kuznetsova et al.,[25] Fox and Weisberg,[26] R Core Team[27]	https://cran.r-project.org/

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Sissel Sjöberg (Sissel.Sjoberg@biol.lu.se).

Materials availability

This study did not generate any new unique materials.

Data and code availability

- Raw data from Figures 1 and 2 and all analyses were deposited at Mendeley Data, https://doi.org/10.17632/k7m3gnswdj.1
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Study systems

Adult great reed warblers were caught and equipped with multisensor data loggers (MDLs) at Lake Kvismaren and nearby Segersjö in south central Sweden (59°10′N, 15°24′E) during the breeding seasons 2017–2019. In total 63 MDLs were attached to the back of the birds using leg-loop harness in 2017–2018, and 29 out of these were retrieved the following year (46% return rate which is even higher than the 35% return rate (23 out of 65) for birds without loggers in these years). The mass of MDLs plus attaching material was approximately 1.1 g (< 4% of the normal body mass of the great reed warbler during the breeding season). Due to technical problems, only 14 MDLs contained altitude data, and these were used for further analyses (see below).

The great snipes belonged to a breeding population in the Storulvån area in Jämtland, west central Sweden (appr. 63.17°N, 12.38°E[2,28]). In 2015–2019, 107 adult great snipe males were equipped with MDLs on a ring on their tibia, and 36 out of these were retrapped in the following one to three years (34% return rate which is similar to birds with only a metal ring [31%]). The mass of the MDLs plus attaching material was approximately 1.4–1.7 g (~1% of the normal body mass of the great snipes during breeding season). Four birds had lost their loggers, and technical problems with the barometric pressure sensor and/or time keeping issues further reduced the sample to 14 MDLs that included altitude data for at least one of the long flights (autumn, spring or the winter movement in Africa[2]). For birds carrying a MDL that recorded data for several years either the most complete annual dataset, or data from only the first year, was included in the analyses.

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METHOD DETAILS

Multisensor data loggers (MDLs)

The MDLs used in this study included an accelerometer for activity measures, a temperature compensated barometric pressure sensor (recording both temperature and barometric pressure), a light-level sensor, a real-time clock and memory.[5,6] All data presented in this study are from the same flights, and recorded with the same MDLs, that were used to describe the diel cycle in flight altitudes in the previous studies of great reed warblers[1] and great snipes.[2] For details about activity measures, inclusion and validation of data, and calculations of flight durations, see.[1,2,5]

Barometric pressure and logger temperature were measured once every hour with an absolute accuracy of ca ± 1 hPa at 0°C–40°C (equivalent to ca ± 8 m), and ± 1°C (± 0.5°C around 25°C), respectively. The sensor used was Bosch Sensortech BMP280, an individually factory pre-calibrated digital sensor. Flight altitudes were calculated from the barometric pressure data using the International Standard Atmosphere model (SA; International Organization for Standardization 1975: ISO 2533:1975).[6] Depending on variations in local atmospheric conditions, SA height can potentially differ by ± 10% from true (geopotential) altitudes.[6] For both great reed warblers and great snipes the SA altitudes are probably slightly underestimated compared to true altitudes.[1,2] Expected ambient air temperature data (used as an illustration in Figure 2) were calculated from the SA flight altitudes of the birds, using the International Standard Atmosphere model (SA; International Organization for Standardization 1975: ISO 2533:1975)[6] which assumes a temperature decrease of 6.5°C / km altitude and 15.0°C as the temperature at sea level.

QUANTIFICATION AND STATISTICAL ANALYSIS

Data treatment and statistical analysis

To analyze logger temperatures in relation to flight altitudes during day and night, all data were categorized as day (08:00–14:00 UTC), night (20:00–02:00 UTC) and intermediate (all other times; excluded from analyses) periods. Only flights that have both diurnal and nocturnal data were included in the analyses. This resulted in a sample of 26 flights from 14 individual great reed warblers (10 autumn and 16 spring flights) and 25 flights from 14 individual great snipes (13 autumn, 7 winter and 5 spring flights).

General Linear Mixed Effects Models (GLMMs) were calculated separately for the two species to investigate differences in logger temperature between day and night at the same altitudes with individual and day/night number being used as random factor. The models were restricted to flight above 2,000 m asl and below 5,000 m asl, since at < 2000 m asl there may be significant diel variation in ambient temperature[9] and at > 5000 m asl there was no overlapping data between diurnal and nocturnal flights (for the great reed warbler model only data from 13 individuals was included in the model as one individual lacked data in the interval 2,000–5,000 m asl). As there is no reason to expect a difference in the relationship between altitude and logger temperature between different seasons, or between flights from autumn, spring, and winter (only great snipes), seasons were combined in the analyses. All models were implemented using the package lmerTest[25] and visualized using the package effects[26] for R 3.4.4.[25,27]